Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

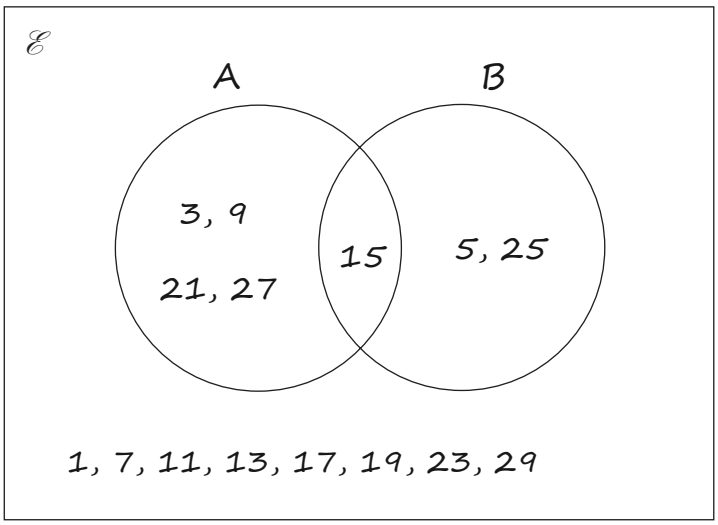
1 $\mathscr{E}$ = {odd numbers less than 30}
A = {3, 9, 15, 21, 27}
B = {5, 15, 25}

(a) Complete the Venn diagram to represent this information.

1 mark
Label A and B

1 mark
Only 15
in the overlap

1 mark
Correct entries
in another region



1 mark
All numbers
correctly placed

(4)

A number is chosen at random from the universal set, $\mathscr{E}$.

(b) What is the probability that the number is in the set $A \cup B$?

The union means A or B, including the overlap.

7 numbers in the union; 15 in the universal set.

1 mark
Use a count from the Venn diagram

$\frac{7}{15}$

1 mark
Correct probability

(2)

(Total for Question 1 is 6 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA